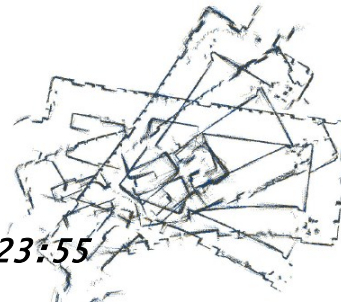


EE 585 Probabilistic Robotics

Homework #3 (Individual work)

Given: Nov 19, 2025 (Wed), Due: Dec 05, 2025 (Fri) 23:55



General Information About the Homeworks:

Homeworks are to be prepared in Electronic Form composed of an Adobe pdf format homework document "Surname_HW1.pdf" and associated Matlab/Python files (if applicable) having name specified in the homework. All should be packed in a zip file having Surname_hw1.zip and uploaded to the appropriate assignment entry in OdtuClass. If required, the document should give all derivations, mathematical solutions, Matlab/Python figures and comments on your results. All equations must be typeset in a legible and understandable form. The organization and neatness of the homework is an important consideration in grading.

The electronic document should be accompanied by the necessary Matlab m-files (only data files and plotting functions if Python is used) that can be run from a clean Matlab session (no previously defined variables) and provide all the necessary output. The programs should indicate on the command window what they are in the process of doing.

For this homework you will implement the solutions in m-files named q1_sol.m, q2_sol.m, and q3_sol.m. These m-files should put out all necessary comments and draw the figures that are also integrated into your homework document.

Q1: In this question, you will implement a Kalman filter to estimate the state of a simple dynamic system. You will need to review the "Gaussian Filters" chapter of your textbook, and the lecture slides on this topic.

Consider a "car" with linear dynamics moving in a linear environment (i.e., on a 1D line). Assume $\Delta t=1$ for simplicity. The position of the car at time t is given by x_t . Its velocity is $\dot{x}_t$ and its acceleration is $\ddot{x}_t$. Suppose the acceleration is set randomly at each point in time, according to a Gaussian with zero mean and covariance $\sigma^2=1$.

- (a) Describe the minimal state vector for the Kalman filter so that the resulting system respects the Markov assumption.
- (b) "Simulate the true motion". I.e, generate and plot (use connected plots) 3 example instances of the true state trajectory (use different color and plot on the same plot using "hold on"). Note that since the acceleration is random, each instance will result in a different true trajectory.
- (c) For your state vector definition, design the state-transition density $p(x_t | u_t, x_{t-1})$ by defining the matrices **A**, **B** and noise covariance matrix **R**.

- (d) Implement the state prediction step of the Kalman filter. Assuming that we know that at time $t=0$ we have $x_0 = \dot{x}_0 = \ddot{x}_0 = 0$, compute the state prediction density (i.e. the mean vectors and covariance matrices) for times $t=1,2,3,4,5$.
- (e) For each value of t , plot the joint state prediction density over x and $\dot{x}$ in a diagram where x is the horizontal and $\dot{x}$ is the vertical axis. For each posterior, you are asked to plot an *uncertainty ellipse*, which is the ellipse defined by points that are one standard deviation σ away from the mean. You can create those ellipses by analyzing the eigenvalues of the covariance matrix.
- (f) what happens to the correlation between x_t and $\dot{x}_t$ as $t \rightarrow \infty$? Illustrate by plotting a number of uncertainty ellipses for large t .

Q2: Now, we will simulate sensor measurements from our simple car dynamic system and observe the effects of these measurements on the state estimation by the Kalman filter. Suppose that at time t , we can receive a noisy observation of the position x . In expectation, our sensor measures the true location. However, this measurement is corrupted by Gaussian noise with $\sigma^2=10$.

- (a) Define the probabilistic measurement model $p(z_t | x_t)$ by specifying the matrix C and noise covariance matrix Q .
- (b) Implement the measurement update step of the Kalman filter. Suppose at time $t=5$, we make a measurement $z=5$. State the parameters of the Gaussian estimate before and after incorporating the measurement (measurement update step of the KF).
- (c) Plot the state *uncertainty ellipses* before and after integrating the measurement (as described in the previous question).
- (d) what would have happened if we were to make measurements for time steps $t=1,2,3,4,5$ and integrate them with the KF. Illustrate by plotting the uncertainty ellipses of the state estimate after each full KF iteration.

Q3: We will now implement the same state estimation task with the Particle filter approach. We will assume the same linear-Gaussian motion and measurement models as in Q1 and Q2.

- (a) Implement the functions `low_variance_sampler`, `sample_motion_model` and `measurement_model` as required by the Particle filter algorithm.
- (b) Implement the Particle filter algorithm.
- (c) Repeat (d) and (e) of Q1 by considering $N=1000$ particles. Instead of the uncertainty ellipse based

on Gaussian parameters of the estimate, you will plot your particles. Make sure that particles plotted for different t can be distinguished. You may use different colors, examine some plots from your book to see some examples of appropriate ways of plotting.

- (d) Repeat (c) and (d) of Q2. Again make sure that we can distinguish particles from different time steps as well as those before and after integrating the measurement.
- (e) Attempt a complexity comparison of KF and Particle Filter by using the Matlab execution time measurement functions. E.g., You may attempt measuring one full iteration of KF and PF with $N=1000$.

Good luck.